Study of **M**obile **A**d hoc **Net**work routing protocols based on NS3 and SUMO simulation

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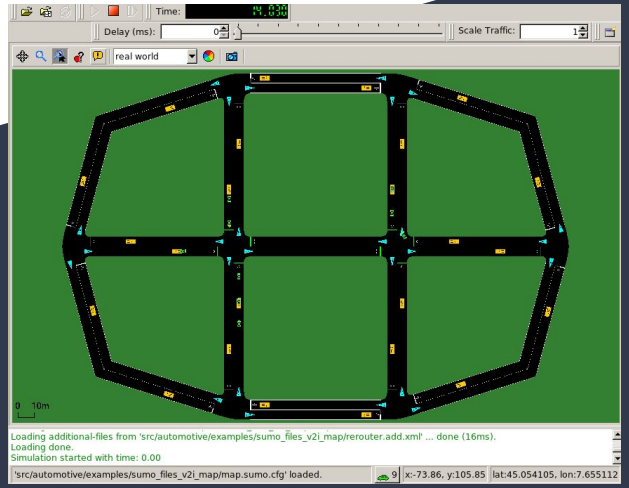


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Introduction

MANETs : Autonomous systems of mobile devices connected wirelessly without fixed infrastructure

- “Self-configuring”
- Each node acts as a router
- Applications : Emergency response, smart cities…
- Challenges : maintain communications with scalability, resource constraints…

2 types of protocols :

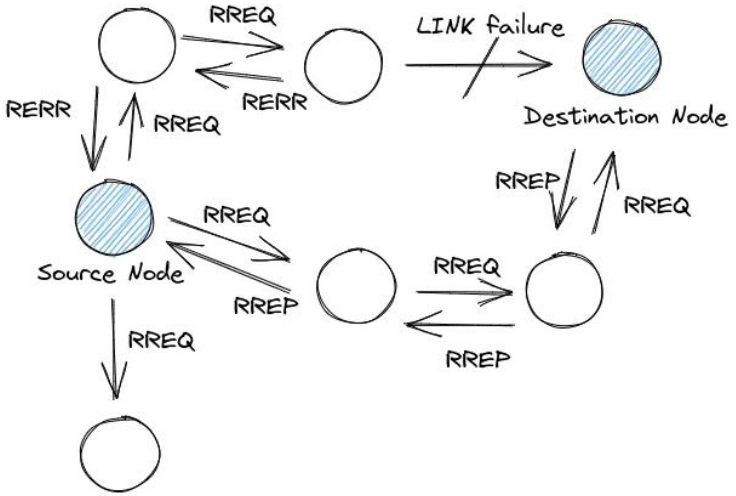
- **Proactive** : table-driven protocols, maintain up-to-date routing tables through periodic updates (DSDV, OLSR).Require high P and B.
- **Reactive** : on-demand protocols, establish routes only when they're needed for data transmission (AODV, DSR). Require low P and B.

MANETs protocols ⇒ conditions of changing topology and variable wireless link quality

AODV & DSR

AODV:
Ad-hoc On-demand Distance Vector Routing Protocol

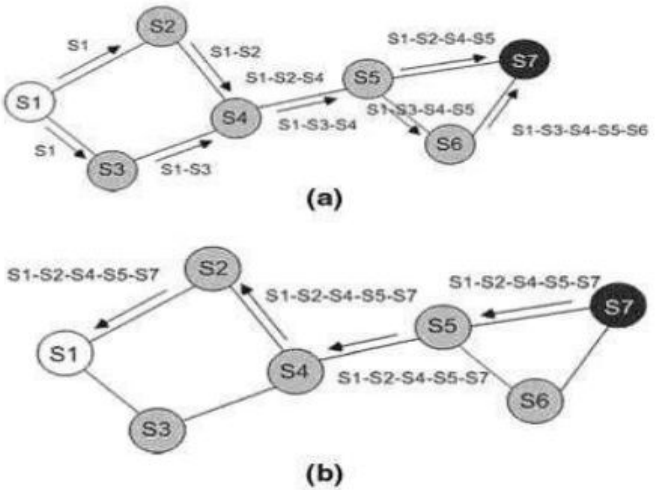
- Nodes discover routes using RREQ, RREP, RERR, only when required
- Routing tables with limited lifetime (next hop)
- One route possible
- Using sequence numbers (route freshness)



AODV & DSR

DSR:
Dynamic Source Routing protocol

- The source node determines the entire route to the destination (packet header)
- Node uses a routing cache to store learned routes for future use
- Multiple routes possible



(a) Route Discovery **(b)** Using route record to send the route reply

AODV & DSR

Parameters	AODV	DSR
Source Routing	No	Yes
Time Relay	Does relay	Doesn't relay
Path Discovered	Single route	Multiple routes
Periodic Broadcast	Yes(HELLO message)	No
Routing Overhead	Less	More
Route Storage Structure	Routing Table	Route Cache
Benefit	Adaptable to highly dynamic network topologies	Multiple Routes
Caching Overhead	Low	High

Key Differences

- **Route Discovery**:
  AODV : each node maintains only next hop
  DSR : RREQ packet stores the entire route
- **Route Maintenance**:
  AODV: Uses route tables (next hop)
  More route rediscovery
  DSR : Uses route caching (complete route)
- **Overhead**:
  AODV: lower packet delay
  DSR: massive delay due to source routing

SUMO

Simulation of Urban MObility (SUMO):

- open-source traffic simulation framework
- models urban traffic, including vehicles, pedestrians, and public transport.

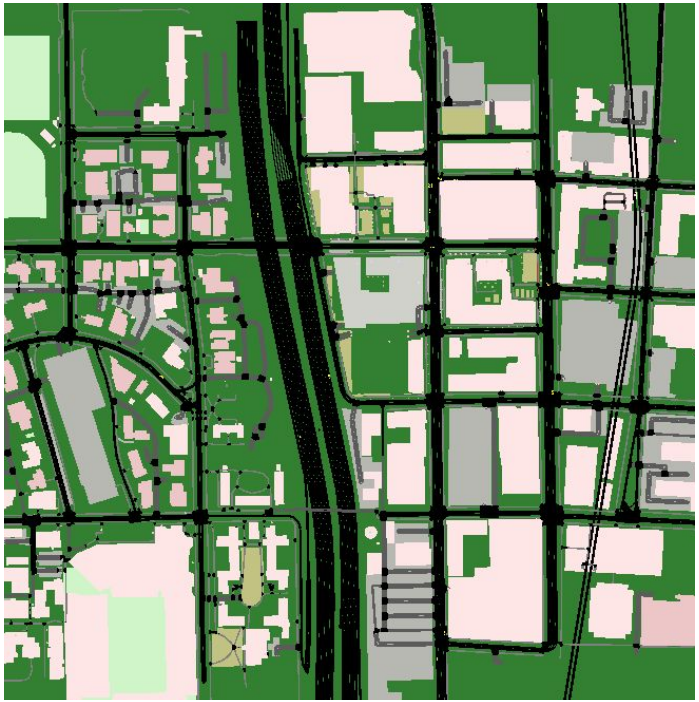
⇒ Realistic mobility patterns for nodes

OSMWebWizard:

- Based on real world data from OpenStreetMap



Simulation implementation



- Parse the SUMO trace file to get the nodes, their positions, movements, exit/entry times
- Set up a different routing helper depending on the chosen protocol
- Onoff application :
  - Intermittent traffic generation
  - UDP packets
- Wifi ad hoc mode : LAN without routers/access points
- Upon entering, start broadcasting to a random active node until one of them leave the network

Performance evaluation

Packet size: 512 bytes

Rate: 2048 bps

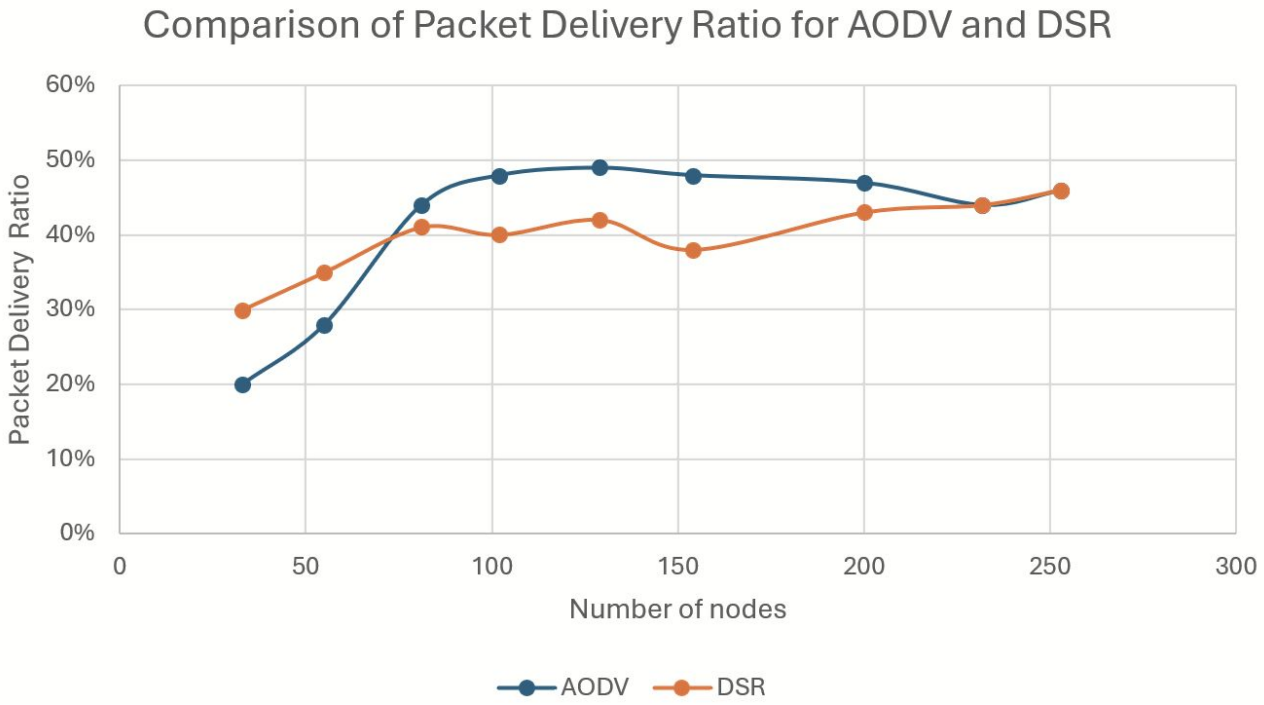
Continuous broadcast

Signal Power: 10 dBm

Wifi IEEE 802.11p

Friis loss model

Simulation Duration: 600 s



Performance evaluation

- **Network Throughput**

Expectation: Similar throughputs between DSR and AODV

- **End to end Delay**

Expectation: DSR ⇒ Very high delay
AODV ⇒ Lower delay

- **Routing/Packet overhead** ⇒ High routing overhead = reduced bandwidth

Expectation: DSR ⇒ Low overhead
AODV ⇒ High overhead

Improvement points

- **Add buildings impact on network**

- **Use different NS3 applications:**
  e.g. BSM (Basic Safety Message) application ⇒ messages exchanged between vehicles and infrastructure in applications related to Intelligent Transportation Systems (ITS) and Cooperative Intelligent Transport Systems

- **Change propagation model**

Possible future works

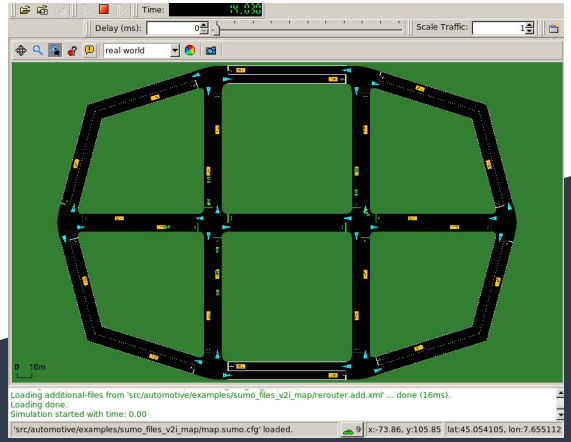
- **Advanced Routing protocols**
  new routing protocols: adaptive, energy-efficient and hybrid routing protocols

- **Investigate security and privacy of such networks**
  what to expect in terms of threats

- **Integration with Emerging technologies**
  autonomous vehicles, disaster response, smart cities.

Thank you for your attention !

Any questions ?



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